

Forecasting key indicators of China's inbound and outbound tourism: optimistic–pessimistic method

Forecasting
China's
tourism
indicators

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Received 20 December 2019

Revised 3 May 2020

28 June 2020

Accepted 1 July 2020

Abstract

Purpose – Tourism industry is a highly complex system surrounded by many uncertainties because of its innumerable connections with other supporting systems. Considering tourism, a grey system, the current study proposes optimistic–pessimistic method (OPM). This technique can aid in improving forecast accuracy of four tourism-related indicators, inbound tourism to China, outbound tourism from China, revenues collected through inbound tourism and expenses incurred on outbound tourism.

Design/methodology/approach – The study integrates OPM into EGM and then using the secondary data collected from the World Bank database, predicts the four tourism-related indicators. The mean absolute percentage error steered the performance of the models.

Findings – One of the main contributions of the study lies in its overall evaluation of one of the major travel and tourism countries of the world in light of four crucial indicators. The study highlights, four tourism-related indicators' recent information, contains more valuable information about the future.

Originality/value – OPM represents a novel application of concept of whitenization of interval grey number in grey forecasting theory.

Keywords Traveling, Inbound tourism, Outbound tourism, Grey forecasting model, Optimistic-Pessimistic Method

Paper type Research paper

Nomenclature

EGM (1,1)	Even grey model, first-order differential equation containing one variable	GDP	Gross domestic product
		GM	Grey model for forecasting
		GST	Grey system theory

This work was supported by a project of the National Natural Science Foundation of China entitled “Research on network of reliability growth of complex equipment under the background of collaborative development” (71671091). It is also supported by a joint project of both the NSFC and the Royal Society of the UK entitled “On grey dynamic scheduling model of complex product based on sensing information of Internet of things” (71811530338), a project of the Leverhulme Trust International Network entitled “Grey Systems and Its Applications” (IN-2014–020). At the same time, the authors would like to acknowledge the partial support of the Fundamental Research Funds for the Central Universities of China (NC2019003) and support of a project of Intelligence Introduction Base of the Ministry of Science and Technology (G20190010178).



Grey Systems: Theory and
Application
© Emerald Publishing Limited
2043-9377
DOI 10.1108/GS-12-2019-0064

GS	MAPE	Mean absolute percentage error	USD	US Dollars
	OPM	optimistic–pessimistic method	WTCF	World Tourism Cities Federation
		(O/P Method)	WTTC	World Travel and Tourism Council
	O/P-EGM	O/P Method– based EGM (1,1)		

1. Introduction

As growth in international tourism is slowing down in mature destinations (e.g. Northern and Western Europe and North America) and is expected to increase in emerging destinations at twice the rate of those in advanced nations, tourism in emerging markets is drawing increasing attention (Claveria, 2016). China has witnessed substantial growth of international tourism over the past twenty years and today has become a key originator and recipient of tourists (Sun *et al.*, 2016). Today, China has emerged as one of the “major Travel and Tourism economies” of the world (WTTC, 2017a), and not only China itself is reaping benefit from international tourists so do countries China’s “generous” travellers chose to visit. Today tourism has become a vital engine of socio-economic development for countries through export revenues, employment opportunities and infrastructure development (World Bank, 2018). In 2012, China became the most prominent international tourism spender for the first time, setting a record \$102bn in tourism expenditure and at the same time got the number 1 outbound tourism status in the world (Chen, 2018; Chen *et al.*, 2017). In 2016, international inbound tourists spent 114.109bn USD in China, according to World Bank statistics. As a globally traded service, inbound tourism is recognized for being one of the leading trade categories in the world. In the decade from 1994 to 2004, outbound international travellers from China increased by 17% annually, and in the period 2010 to 2014 China’s outbound tourists increased at an average annual rate of 18% (WTCF, 2018; Keating and Kriz, 2008). Considering the proportion of Chinese outbound tourists to be small as compared to its population, comprising a continually expanding middle class, there is an enormous growth potential for Chinese outbound tourism with positive consequences for the destined countries (Keating and Kriz, 2008). In 2017, 130.5m outbound tourists from China spent \$258bn, with a growth of 7% (outbound tourism) and 5% (expenditures) (WTCF, 2018). These statistics show the momentum of Chinese tourism industry, which is bigger than the economies of several countries. It also shows tremendous potential that China as a destination and source of tourists possess. Also, considering importance of international tourism for China and its economy and significance of Chinese outbound tourists for countries they visit, its forecasting has both policy and social implications for the concerned governments and other stakeholders like travel agencies, airline industries, hospitality industry and even general public.

From 2016 to 2017 even though amount of foreign visitor exports (spending) rose from \$119.7bn to \$125.3bn however the proportion was a reduction in 5.3% to 5.2% of total exports, with foreign-visitor spending reducing from 18% to 13% (WTTC, 2018). Further, even though total value of tourism trade experienced a geometric increase since 1978, however in 2009 China entered into a deficit period of tourism foreign exchange income, and the deficit period is still not over (Yan, 2017). Thus, development in Chinese tourism is accompanied by a continuous and expanding deficit in tourism balance of trade (Dai *et al.*, 2017). Albeit, the overall picture of tourism industry of China is good, and the country is on its way to becoming the most attractive tourist destination in upcoming years. However, considering the existing and potential challenges, there is a continuous need to re-study and research Chinese tourism sector and its performance especially when China has become one of the most interesting and widely discussed countries in the world not only because of its economy, global influence, technology, mega-projects, food and culture but also because of its tourism-related policies and its consequences. In order to make the right policy and

investment decisions, the importance of empirical evidence is immense (WTTC, 2018). Thus, the need to empirically study Chinese tourism sector's current status and its future performance is growing which can also be manifested from increasing numbers of research publications on Chinese tourism-related phenomenon (see, e.g. Cheng and Zhang, 2019; Gao *et al.*, 2019; Huang and Chen, 2016; Ruhanen *et al.*, 2015). Roles of support vector regression, artificial neural networks and fuzzy approaches are well known (Wan and Song, 2018) for tourism-related forecasting; however, the deployment of grey models on tourism-related predictions is still in infancy. Furthermore, given significance of China as a major tourist destination and uncertainties surrounding tourist behaviours when emerging destinations are outpacing advanced destinations at a higher speed in terms of international tourism growth, it is crucial for concerned organizations to study the behaviour of Chinese tourism through an approach that is better equipped to handle uncertain and complex systems like tourism industry, and is not unfamiliar with the Chinese environment as well.

Considering tourism industry a huge, comprehensive, complex, volatile (sensitive) and uncertain system (Sun *et al.*, 2016; Xu, 2008; Gössling and Hall, 2006), comprising multiple interlinked sub-systems (businesses/sectors/supply chains) that may differ in "nature, operations, and external environment" (Marais *et al.*, 2017), which becomes more comprehensive, as well as uncertain, in the Chinese context because of number of factors such as higher number of tourist destinations, variety of seasons, higher considerations like climate change and pollution, increasing ageing population, parallel systems of habits (e.g. eating and cuisines) and cultures (consider, a nation comprising 56 ethnic groups), parallel systems of payments (online versus cash), parallel systems of governances ("one country, two systems"), a mix of socialistic and market-oriented administrative policies and diverse actors/stakeholders involved, the current study aims to forecast inbound tourism to China, outbound tourism from China and related receipts and expenditures. Based on the data from 2006–2016, the predictions for the next 10 years have been made using a grey prediction model, Even GM(1,1). While discussing the model GM(1,1), which is usually synonymous to model Even GM(1,1) in the literature, Sun *et al.* (2016) argued that although the model has been broadly studied, it may have unsatisfactory curve-fitting effects for data revealing significant randomness; thus it is essential to propose new models to improve its forecast accuracy. Therefore, while making the objective of the current study manifold, a novel and convenient approach, optimistic–pessimistic (O/P) method (OPM), has also been proposed and successfully tested to improve the forecast accuracy of original grey model.

The structure of the paper is as follows: After the introduction section, there is the section on background. Within it, a review of literature justifying the appropriateness of grey forecasting models (GMs) for tourism forecasts is also presented. It is followed by the section on research methodology where the data collection and analysis methodologies are briefed. The study reveals that tourism data available in major databases contain uncertainty, and thus deployment of the models of grey system theory, an emerging methodology to analyse data containing uncertainty, was a sound move.

2. Background

2.1 Chinese tourism industry

As one of the biggest economic industries of the world, travel and tourism, which supports one-tenth jobs on the globe, is a dynamic driver of employment opportunity and can absorb surplus labour in a country (WTTC, 2018; Wei *et al.*, 2013). Tourism industry accounts to around 10% of global GDP and is projected to grow at a rate of 4% per annum, thus eclipsing many other industries (Lenzen *et al.*, 2018; WTTC, 2018). One of the key benefits of tourism for developing countries like China, where unbalanced regional development and increasing regional economic disparities are significant challenges for a balanced national economic growth, is its inherent characteristic to distribute development away from industrial centres

and towards regions that have not been developed (Goh *et al.*, 2015). In post-Mao period, when uneven regional and economic development started showing up, tourism became an important cornerstone of Chinese national policy to support national economy. Since the late 1970s, China's appreciation of tourism has become a moderator for economic development and modernization that allowed Chinese tourism sector to develop rapidly and making it the largest sector in service industry (Wei *et al.*, 2013; Keating and Kriz, 2008). "The success of the economic reforms in China, which has resulted in fast economic growth has also been the main cause of rapid tourism growth in China," states Witt and Turner (2002). International tourist arrivals surged from 1.18m in 1978 to 106.6m in 2011, and foreign exchange revenues from inbound tourism rose from \$263mn in 1978 to \$48.5bn in 2011 and then \$56.9bn in 2014 (Keyim, 2017; Yan, 2017). Overall, from 1980 to 2006, inbound tourist arrival to China and resulting tourism receipts grew at a rate of 11 and 14% per annum, respectively (Keyim, 2017). Inbound tourism is an internationally traded service and a major category of trade, and it has an effect on an economy like that of exports. In 2016, international inbound tourists spent 114.109bn USD in China, according to World Bank statistics.

Tourism statistics place China in a very lucrative position in the world both as a source and destination country of tourists. Thus, not only macro-factors are being studied, but micro-factors have also sought scholars' attention to understand the market better. For example, scholars have even begun studying behaviours of Chinese tourists to serve them better. Cheng and Zhang (2019) investigated the Airbnb hosts' experience with Chinese outbound tourists. Ruhanen *et al.* (2015) studied Chinese tourists' perceptions and motivations for a particular (e.g. Australian) tourist destination. In this new era of tourism, where tourism can drive a country's economy, no wise country affords to refuse tourists who are well-known for their generosity and spending. At the same time, irrespective of the fact that what people in West may think of China, rising number of foreigners visiting China each year demonstrates their curiosity in the country that houses almost everything – from beautiful mountains to well-preserved historic sites, from cold desert nights to hot springs and from lush green forests to high rising skyscrapers with a sense of safety and security at every mile.

Further, the efforts China's tourism bodies are putting in to manage these cultural, historical and natural sites are not going astray. Zha and Li (2017) notes that "the tremendous progress in the Chinese tourism sector has become one of the key factors fuelling the growth and advancement of the Chinese economy." Every year the World Travel and Tourism Council (WTTC) disseminates a report of tourism performance of countries. The report is considered authoritative among stakeholders associated with tourism industries all over the world. According to their reports (WTTC, 2017a, 2018), the *direct* contribution of travel and tourism in China's GDP rose from USD275.2bn (2016) to USD402.3bn (2017) with 2.5% (2016) to 3.3% (2017) of total GDP. The *total* contribution, which includes direct and indirect contributions, of travel and tourism to GDP rose from USD1000.7bn (2016) to USD1349.3bn (2017), with 9% (2016) to 11.0% (2017) of GDP. These figures make tourism and travel not only a vital part of the Chinese economy but also signify the importance of China as a worthy place for visitors and travellers. These figures also signify China, which is already the world's second-largest economy, as one of the major travel and tourism economies as well.

2.2 Tourism industry as a grey system

Tourism research has become a multi-disciplinary field (Faulkner and Goeldner, 1998) as no single theory can explain the tourism system and phenomenon independently because of the relation of tourism systems and tourism industry with almost any other system and industry in the world. For example, the disease of COVID-19 that emerged as a healthcare crisis did not take too long in evolving to the travelling/tourism crisis, affecting not only every stakeholder in travelling and tourism industries but also in the closely-linked industries. Thus, studying

tourism in the context of networks, chaos and complexities, and uncertainty (Wu *et al.*, 2019; Baggio and Sainaghi, 2011, 2016; Pavlovich, 2003, 2014; Xu, 2008; Zahra and Ryan, 2007; Gössling and Hall, 2006) is an important issue. Travelling is a phenomenon that surrounds much uncertainty for travellers who leave their comfort zones to go abroad. However, if one looks at tourism industry on the whole one may see a highly complex, open and flexible (self-organizing) system continuously striving to achieve equilibrium in the wake of rising complexities across all businesses directly or indirectly connected with national or international tourism infrastructures. At least since the late 20th-century tourism as a complex, chaotic and uncertain system is getting exceeding attention by the scholars. Faulkner and Russell (1997) studied chaos and complexity in the tourism system. Zahra and Ryan (2007) discussed complexity in the tourism system and acknowledged the difficulty in predicting the future of the system, in light of chaos and complexity theory. Xu (2008) forecasted domestic tourism in Hubei using grey system theory. A system might begin from an ultimately ordered and stable state and develop to states characterized by a complex behaviour, or even to completely chaotic situations where long-term forecasts are impossible because uncertainty in the accurate determination of its initial state evolves exponentially in time (Sainaghi and Baggio, 2017). Therefore, in the current study, tourism has been viewed as an open system, which is both complex and uncertain and continuously interacts with other systems in its surrounding. However, irrespective of amount of uncertainty and unpredictability surrounding this system it cannot be classified as a *black* system, a system with entirely unknown information. At the same times, these complexities and uncertainties restrict us from considering it as a *white* system, a system with entirely known information. Thus, considering the partially known and partially unknown information associated with it, one can view tourism as a *grey* system. This perspective does not negate the existing theories and their perspectives but rather complements them to better understand the tourism system.

2.3 Tourism forecasts and grey prediction models

Tourism industry is a hugely complex system, which is a set of multiple sub-systems (or, businesses) like tour guiding, passenger transportation, retailing, hoteling, catering, event management and several support and leisure services (Marais *et al.*, 2017). As most of tourism products (e.g. airline seats, hotel rooms and car rentals) are perishable, and cannot be inventoried, thus considering the fact that fluctuating demands cannot be met with a stable supply of resources, equipment and workforce, the need for reliable forecasting of tourist arrivals and demands for accurate decision-making becomes a fundamental need (Hadavandi *et al.*, 2011; Hsu and Wang, 2008). Therefore, there is no escape from forecasts, and if forecasts are accurate, then it is a blessing otherwise lamentation! Accurate forecasts can also minimize need for risk and uncertainty management, and if a risk and uncertainty management is inbuilt in a forecasting technique, benefits can be manifold.

Knowing the fact that tourism is a very comprehensive industry representing a very complicated, uncertain and huge system (Xu, 2008), which comprises dozens of sub-systems, tourism industry-related forecasts need a non-traditional approach like grey system theory that allows the tourism-related forecasts to be nearer to reality. Grey forecasting models (GMs) are an integral part of grey system theory, which is an intelligent, multifaceted field, which was pioneered by Deng (1982) to handle problems associated with poorly understood (uncertain) systems (Mahmoudi *et al.*, 2020). One of the earliest studies using a grey model to forecast tourism demand was executed by Wang (2004). He argued that in order to obtain low forecast errors from autoregressive-integrated moving average (ARIMA) and neural network models one constraint is necessity of a large sample, which is difficult to obtain in the environment that is dramatically changing in a short period. Therefore, he preferred grey system theory and fuzzy theory for prediction of tourists' arrival to Taiwan from Hong Kong,

Germany and the United States. Later several scholars used grey models independently or with other models in a tourism environment, (see, e.g. [Huang et al. \(2004\)](#), [Liu et al. \(2008\)](#), [Hsu and Wang \(2008\)](#), [Xu \(2008\)](#), [Lin et al. \(2009\)](#), [Chen and Qiu \(2009\)](#), [Huang et al. \(2009\)](#) and [Kan et al. \(2010\)](#)). Among the recent studies, [Liu et al. \(2014\)](#) used an optimized GM (1,1) model to forecast the number of domestic tourists to Zhejiang. [Pirthee \(2017\)](#) used two grey models to predict tourist arrival to Mauritius. [Ma, Liu and Wang \(2019\)](#) used a non-linear multi-variate grey Bernoulli model to forecast tourist income of China.

Here, it should be noted that currently many forecasting methods are in use for tourism forecast; however, they have different limitations. For example, an ARIMA model is usually used for time series forecasting where historic times series data is used to construct a mathematical function ([Wang, 2004](#)), which is being used to predict future data values. However, in order to reduce forecasting error, ARIMA and other time series models require a large number of data, which is not always and readily available for time series data when the environment is dynamically changing with time or cost of data collection is not affordable ([Sun et al., 2016](#); [Wang, 2004](#)). To overcome these issues, researchers proposed neural networks-based models; however, they require a larger sample size in order to get reliable forecasts. On the other hand, GMs are neither subject to large sample size nor any particular distribution of data ([Liu et al., 2020](#); [Javed and Liu, 2018b, 2019](#); [Ng and Deng, 1995](#)). [Hsu and Wang \(2008\)](#) in their study on tourism demand forecasting also concluded that “in situations involving little sampling data, the grey model is superior to other traditional forecasting models.” [Liu and Liu \(2010\)](#) in their study on predicting red tourism through model GM(1,1) also stated that application of grey models could solve the problem of insufficient and uncertain data even when tourism growth is not stable. In the [Table 1](#), adapted from [Dang et al. \(2016\)](#), an overview of the comparative analysis of GMs and other forecasting models has been presented.

3. Methodology

3.1 Data collection and analysis techniques

Secondary data concerning international tourism (receipts; the number of arrivals; the number of departures; expenditures) were collected from the [World Bank \(2018\)](#) for the period 2006–2016. Even model GM (1,1) was used for the data analysis and forecasting. EGM was executed at the Chinese language *Grey System Modelling* software. For O/P-EGM, Microsoft Excel was also used. For the exponential regression analysis, Microsoft Excel was used.

Here the current study wants to offer a very serious proposition. The data disseminated by an organization (or databank) considered unreliable may be unreliable; however, the data disseminated by an organization (or databank) considered reliable may not necessarily reliable. It happens because of uncertainty in the data that can make reliability of the data questionable ([Javed, 2019](#)). For example, on 11 November 2018, the database of World Bank ([data.worldbank.org](#)) was showing that “International tourism, expenditures (current US\$)” were 261,129,000,000\$ in 2017. However, on 10 July 2019, the same database was revealing the expenditures to be 250,112,000,000\$ (while the figure for 2018 was 257,733,000,000\$). During the Pandemic I, the

Table 1.
Historical data
requirement of
different models

Forecasting models	Data requirement	Time horizon for forecasting
Box–Jenkins	50	Long
Moving average	2 to 30	Short
Simple exponential smoothing	5 to 10	Short
Neural network	Large number	Short
Trend models	10 to 20	Short, medium
Casual regression models	10 observations per independent variable	Short, medium, long
Grey models	4	Short, medium, long

COVID-19 pandemic, the international community witnessed several instances on the global stage when different parties revisited their databases on different occasions to reflect the reality more accurately. Other than underscoring reliability of data as a function of time these facts signify the relevance, and significance, of latest information over older one. Thus, if a forecasting methodology is unable to consider uncertainty in data, accuracy of a forecast can be tainted. An approach like grey system theory, which was primarily introduced to deal with problems containing uncertain data or poor information, is equally suitable when data (or information) do not apparently contain uncertainty. When new information is available, value of old information may change (degrade) prompting database administrators to revise their earlier datasets.

3.2 The model Even (1,1)

Grey system theory (GST) is an interdisciplinary and intelligent approach to analyse data associated with a “grey system”, an uncertain system with incomplete or partially known information (Mahmoudi *et al.*, 2019; Javed *et al.*, 2018; Javed and Liu, 2018a). One of the key advantages of GST and its models is superior decision-making and forecasting even when a sample size is small (Ofosu-Adarkwa *et al.*, 2020; Javed *et al.*, 2019; Carboni and Russu, 2014). For GMs, the sample size can be as small as four values (see, Table 1). GMs, as time-series analysis techniques, have been used in various fields only with partly known distribution information (Wei *et al.*, 2018). GMs assume that original time-series is partially quasi-exponential (Wei *et al.*, 2020). Deployment of different kinds of GMs in tourism forecasting is not unknown in the literature (see, e.g. Pirthee, 2017; Liu *et al.*, 2014). There are four basic forms of model GM (1,1) from which Even GM (1,1) is the most popular one and is almost synonymous to GM (1,1) in communications of the grey systems scholars (Javed and Liu, 2018b; Liu *et al.*, 2017), i.e. when the grey systems scholars refer “GM (1,1)” they usually mean Even GM (1,1). In the following part, the model EGM, adapted from Liu *et al.* (2017) and Liu and Yang (2017), is summarized.

Let actual data sequence is $X^{(0)} = (x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(n))$ and its 1-AGO sequence is $X^{(1)} = (x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(n))$ where, $x^{(1)}(k) = \sum_{i=1}^k x^{(0)}(i)$, $k = 1, 2, \dots, n$. Even form of GM (1,1) will be defined as, $x^{(0)}(k) + az^{(1)}(k) = b$ where, $z^{(1)}(k) = \frac{x^{(1)}(k) + x^{(1)}(k-1)}{2}$. Parameter vector $\hat{v} = [a, b]^T$ can be estimated through the least squares method, i.e.

$$\hat{v} = [a, b]^T = [B^T B]^{-1} B^T Y$$

where,

$$B = \begin{bmatrix} -z^{(1)}(2) & 1 \\ -z^{(1)}(3) & 1 \\ \vdots & \vdots \\ -z^{(1)}(n) & 1 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} x^{(0)}(2) \\ x^{(0)}(3) \\ \vdots \\ x^{(0)}(n) \end{bmatrix}$$

The time-response function of the model is expressed as

$$\hat{x}^{(1)}(k) = \left(x^{(0)}(1) - \frac{b}{a} \right) e^{-a(k-1)} + \frac{b}{a}, \quad k = 1, 2, \dots, n$$

The regressive reduction formula is expressed as

$$\hat{x}^{(0)}(k) = \hat{x}^{(1)}(k) - \hat{x}^{(1)}(k-1), \quad k = 1, 2, \dots, n$$

And the time-response function of $X^{(0)}$ is given by

$$\hat{x}^{(0)}(k) = (1 - e^a) \left(x^{(0)}(1) - \frac{b}{a} \right) e^{-a(k-1)}, \quad k = 1, 2, \dots, n$$

Here it is worth mentioning that in most grey models the first value of a simulated data sequence is the first value of actual data (see, e.g. [Xie and Liu, 2005](#)). Even GM (1,1) exploits first-order differential equation to describe the variable to be forecast ([Sun et al., 2016](#)) and is suitable for non-exponential increasing data sequence ([Liu et al., 2017](#), p. 153). For complete details on Even GM (1,1), its parameters and characteristics, [Liu et al. \(2017\)](#) and [Liu and Yang \(2017\)](#) can be referred.

3.3 Optimistic–Pessimistic Method (OPM)

Here it should be noted that integration of two different approaches/algorithms to boost the performance of a model or to enhance forecasting accuracy of a model is well-known in applied studies. [Li, Yang and Li \(2018\)](#) argued that “a single prediction model cannot accurately fit a data sequence with large fluctuations” followed by proposing integration of linear regression and discrete GM (1,1). Different scholars tried to improve performance of grey models in different ways. One of the earliest studies to provide solid grounds for improvement of GM (1,1) was by [Feng \(1997\)](#). Later different scholars tried to improve grey models in various ways, see, e.g. [Wang \(2004\)](#), [Hsu \(2010\)](#), [Shan et al. \(2012\)](#) and [Huang \(2012\)](#). [Zhang et al. \(2019\)](#) discussed the possibility of improving accuracy of GM (1,1) model through various optimization methods, which includes “the optimization of both parameter estimation method and initial condition for GM (1,1) model, the optimization of combination with other model and the function transformation of the modelling sequence.” However, these optimization methods require small to extensive tailoring in the original algorithm of grey model. The method proposed in the current study is simple and can be applied with confidence without much tailoring of the original algorithm of grey models.

Further, considering the fact that old information can also increase forecast errors in grey models ([Juan, 2011](#); [Liu et al., 2014](#)), a new method to confirm it has been proposed. The proposed technique can be easily integrated into any forecasting technique, and increase or decrease in forecast accuracy resulting from this integration reveals the extent of influence of old information on a forecast. The proposed OPM reduces forecast error without altering the original algorithm of grey forecasting model. Here, instead of altering the *process* (core algorithm) that yields *output* (forecast), merely *input* (historical data) is altered, i.e. first of all, the input is decomposed and then passed through the same process followed by synthesis of resultant outputs. It allows the integration of OPM in any appropriate forecasting algorithm very convenient while preventing increase in computational cost. Prediction accuracy is achieved by reducing historic data’s *noise*, an inherent property of a piece of information that determines its usefulness (or uselessness) for decision-making (e.g. forecasting) relative to another piece of information. This noise can also be called the *inherent randomness* in a sequence of data representing an observed behaviour of a phenomenon or the *inherent randomness* existing in observational values of the individual variables of a model ([Liu et al., 2017](#), pp. 184–186). It has been well-known that new information contains less noise than old information. One approach of reducing influence of the noise on prediction accuracy is to increase overall influence of new information in predictions. If *pessimistic* data sequence is the data sequence containing both relatively older and relatively recent information, then the study argues, the *optimistic* data sequence contains only recent information. Thus, a method that incorporates both of these sequences intrinsically incorporates greater say of recent information in prediction compared to older information. It is the underlying concept on which OPM is developed.

3.3.1 Axiom - I. The integration of OPM in a forecasting model reveals superior performance only when latest information contains greater insight than older information about future pattern.

3.3.2 Definition. Let's say $X = (x(1), x(2), \dots, x(k), \dots, x(m-2), x(m-1), x(m))$ is the set of historical (actual) data that is used to produce *pessimistic* set of forecasted data $X_F = (x(m+1), x(m+2), \dots, x(m+n))$. Here, $x(m)$ represents last data value of actual data, and $x(m+n)$ represents last forecasted value, e.g. if $x(m)$ is for 2016, then $x(m+n)$ could be any year after 2016.

Let us say $X' = (x(k), \dots, x(m-2), x(m-1), x(m))$ is a subset containing *recent* values of the set X , such that $X' \subset X$. Using this recent subset of X , an *optimistic* forecasted set is $X'_F = (x'(m+1), x'(m+2), \dots, x'(m+n))$. The decision to select $x(k)$ is up to the decision-maker. If priority is on extremely new information, $x(k)$ would move towards $x(m)$, otherwise, it would move away from $x(m)$. Rule of thumb could be a consideration of not more than half of recent historical data for optimistic predictions. The data sets X_F and X'_F represent pessimistic and optimistic predictions, respectively. If one's objective is long-term predictions, then two points should be considered carefully. In OPM-based grey models: (1) long-term forecasts from long-term historical data are more likely to be accurate; (2) long-term forecasts from short-term historical data are less likely to be accurate. Therefore, it is recommended that X' should not be too small; it should contain at least four values, i.e. $m \geq k + 3$. This assumption is in line with minimum data requirement for grey forecasting models (Deng, 2005).

According to OPM, the synthetic set of both optimistic (X'_F) and pessimistic (X_F) forecasted sets is represented as $\tilde{X}_F$ and is given by

$$\tilde{X}_F = (\tilde{x}(m+1), \tilde{x}(m+2), \dots, \tilde{x}(m+n))$$

where,

$$\begin{aligned}\tilde{x}(m+1) &= \alpha x'(m+1) + (1-\alpha) \cdot x(m+1) \\ \tilde{x}(m+2) &= \alpha x'(m+2) + (1-\alpha) \cdot x(m+2) \\ &\dots \\ \tilde{x}(m+n) &= \alpha x'(m+n) + (1-\alpha) \cdot x(m+n)\end{aligned}$$

Similarly, synthetic simulation of actual data $\tilde{X}$ can also be obtained. Here, $\alpha \in [0, 1]$ is the *weighting coefficient*, or positioned coefficient, which can be tailored as per the situation and need. If decision-makers consider pessimistic predictions (X_F) more realistic, then α should be lower ($\alpha < 0.5$), and if they consider optimistic predictions (X'_F) more reliable then α should be greater ($\alpha > 0.5$). In general, $\alpha = 0.5$ can be chosen, provided Axiom – I is satisfying.

From the above discussion, it can be seen that OPM is an application of concept of whitenization of interval grey number (Liu *et al.*, 2017, pp. 30–31) to grey forecasting theory. To summarize the discussion, it is argued that in OPM, “Optimism” implies a preference for new information (recent data) over old information, and “Pessimism” implies a preference for entire information irrespective of the fact information is old or new.

3.4 Performance evaluation of model

Chopra and Meindl (2016, pp. 178) states, “Forecasts are always inaccurate and should thus include both the expected value of the forecast and a measure of forecast error.” Mean absolute percentage error (MAPE) is a popular method of estimating performance of a forecasting model. It also allows comparative analysis of two models, e.g. improved versus original one, where lower MAPE value implies higher forecast accuracy. MAPE, usually reported in percentage, is given by the following formula,

$$\text{MAPE}(\%) = \frac{1}{n} \sum_{k=1}^n \left| \frac{x(k) - \hat{x}(k)}{x(k)} \right| \times 100$$

where, $x(k)$ and $\hat{x}(k)$ are representing actual (expected) observation and forecast value (simulated value obtained through the model), respectively. Effectiveness of MAPE has already been reported in multiple studies involving prediction-making using grey models (see, e.g. [Javed and Liu, 2018b](#); [Pirthee, 2017](#)). Scholars use Lewis's scale of MAPE values as a means to judge accuracy of forecasts ([Ding, 2018](#); [Lee and Shih, 2011](#); [Chen et al., 2003](#)). This scale is given below.

$$\text{MAPE (\%)} = \begin{cases} < 10 & \text{Highly accurate forecast} \\ 10 \sim 20 & \text{Good forecast} \\ 20 \sim 50 & \text{Reasonable forecast} \\ > 50 & \text{Inaccurate forecast} \end{cases}$$

4. Results and discussion

4.1 Pilot testing of OPM-based model and performance appraisal

Before applying OPM-based grey forecasting models on the case under the study, it is worthwhile to measure performance of OPM on the secondary data, available to us from some published studies. For this purpose, a case has been taken from [Chen et al. \(2010\)](#) 's study that presents data of tourists to Taiwan from three different destinations (Hong Kong, USA and Germany) over the period 1989 to 2003. This data was divided into two parts; 1989 to 1998 (ten years) was considered as historical/actual data and 1999 to 2003 (five years) was assumed as a forecast horizon and was left for out-of-sample testing. Complete historical data (1989–1998) was considered as the pessimistic data sequence, which an analyst would usually consider for prediction. A recent subset from this data, 1994 to 1998 data, was taken as the optimistic data sequence, which is prerequisite for executing OPM. MAPE (%) was considered as a performance evaluation tool. As can be seen from the [Table 2](#), OPM-based EGM (1,1) outperformed original EGM (1,1) in most cases. OPM reduced prediction error from 3.82% to 3.12% for Hong Kong tourists, from 2.47% to 1.99% for American tourists and from 2.76% to 2.11% for German tourists. Prediction accuracy on future forecasts is also visible from lower MAPE (%) value for the American and German tourists to Taiwan, and for Hong Kong tourists MAPE (%) values are almost comparable. The results indicate that the integration of OPM in a forecasting model is likely to boost forecast accuracy.

The pilot testing case, as shown in the [Table 2](#), is revealing that based on historic/actual data in all of the cases, OPM has improved accuracy of grey models. Even in the forecasted horizon, in most of the cases, OPM-based predictions are nearer to actual values. Further, both *in-sample forecasting* ($N = 10$) and *out-of-sample forecasting* ($N = 5$) were used to measure the performance of the forecasting model through MAPE (%). A decrease in MAPE (%) value, which is clearly observable in most cases, showed that with the integration of OPM in EGM (1,1), accuracy of the model is very likely to increase for both in-sample and out-of-sample data. Thus, feasibility and effectiveness of OPM are evident.

Here it should be noted that in the entire “in-sample” testing, the integration of OPM into EGM (i.e. O/P-EGM) resulted in improvement in forecast accuracy. However, in the “out-of-sample” testing, in two cases, O/P-EGM-based forecast was relatively more accurate than that of original EGM, and in one case EGM-based forecast was slightly better than that of O/P-EGM. This behaviour is explainable in the light of the Axiom – I, i.e. the integration of OPM in a forecasting method may only reveal superior performance if latest dataset contains greater insight than entire dataset about future projections. Therefore, this pilot testing reveals the data of inbound tourists' data from USA and Germany contain greater uncertainty, and thus, there exists a more considerable variation between new and older information.

Year	Hong Kong to Taiwan			USA to Taiwan			Germany to Taiwan		
	Actual data	EGM (1,1)	O/P-EGM	Actual data	EGM (1,1)	O/P-EGM	Actual data	EGM (1,1)	O/P-EGM
1989	211,804	211,804	211,804	220,594	220,594	220,594	25,002	25,002	25,002
1990	193,544	183,780	183,780	224,915	236,791	236,791	24,320	25,404	25,404
1991	181,765	193,985	193,985	240,375	245,487	245,487	25,798	26,582	26,582
1992	193,523	204,758	204,758	259,145	254,504	254,504	28,969	27,815	27,815
1993	213,953	216,129	216,129	269,110	263,851	263,851	28,644	29,104	29,104
1994	241,775	228,131	234,953	286,713	273,542	280,127	31,334	30,454	30,894
1995	246,747	240,799	244,324	290,138	283,589	285,680	32,944	31,866	32,450
1996	262,585	254,172	255,673	289,900	294,004	294,247	33,914	33,343	33,576
1997	259,664	268,287	267,569	303,634	304,802	303,083	34,660	34,889	34,745
1998	279,905	283,185	280,039	308,407	315,997	312,197	35,343	36,507	35,959
MAPE (%)		3.82%	3.12%		2.47%	1.99%		2.76%	2.11%
1999	319,814	298,911	293,112	317,801	327,603	321,599	34,190	38,200	37,221
2000	361,308	315,511	306,817	359,533	339,635	331,299	34,829	39,971	38,531
2001	392,552	333,032	321,187	348,808	352,110	341,305	33,716	41,824	39,893
2002	456,554	351,526	336,254	377,470	365,042	351,628	33,979	43,763	41,307
2003	323,178	371,047	352,054	272,858	378,449	362,279	28,577	45,792	42,777
MAPE (%)		14.44%	15.38%		10.31%	10.16%		27.91%	21.81%

Note(s): *pessimistic data is from 1989 to 1998 (five years)
***optimistic data is from 1994 to 1998 (five years)
*****Forecasting horizon is from 1999 to 2003 (five years)

4.2 Application on China's tourism indicators

After successful pilot testing of the model O/P-EGM, it was deemed fit for practical use. First of all, the model EGM (1,1) was deployed to forecast the inbound tourism (A) and related receipts (R), and outbound tourism (D) and related expenditures (E), as shown in the [Table 3](#). The discussion is presented in the following sections.

4.2.1 Forecasting inbound tourism to China. The first part (A) of the [Table 3](#) shows the actual data from 2006 to 2016 for inbound tourism (overnight visitors) to mainland China and predicted inbound tourists from 2017 to 2026 using even grey model. All values of inbound and outbound tourism are in thousands (1,000 persons), and revenue and expenditures from inbound and outbound tourism are in hundred thousand (\$100,000), in tables. As far as inbound tourism (overnight visitors) to China is concerned, the even grey model's accuracy was around 97.4%. According to the model EGM (1,1), around 64,363,000 (64.4 million) persons are increasingly expected to travel to China in 2026.

4.2.2 Forecasting outbound tourism. The second part (D) of the [Table 3](#) shows the actual data from 2006 to 2016 for outbound tourism (number of departures) from mainland China to abroad and predicted outbound tourists from 2017 to 2026 using grey models. All values of outbound tourism are in thousand (1,000 persons). As far as outbound tourism from China to other countries is concerned, the even grey model's accuracy was around 95.5%. According to EGM (1,1), around 592,848,000 (592.9m) people are expected to travel to other countries from China in 2026. The trend is increasing. [Deng \(2018\)](#) quoted Jane Sun for predicting the number of potential Chinese outbound tourists to be 240m by 2020. According to our projections, this milestone is expected to be achieved somewhere between 2019 and 2020, and in 2020 Chinese outbound tourists are predicted to exceed 253m.

4.2.3 Forecasting revenue from inbound tourism. The third part (R) of the [Table 3](#) shows the actual data from 2006 to 2016 for revenue from inbound tourism to China and predicted revenue from 2017 to 2026 using grey models. All values of revenue from inbound tourism are in hundred thousand (\$100,000). As far as revenue from inbound tourism to China is concerned, the accuracy of both grey models was comparable. The accuracy in predictions through grey models turned out to be around 92.3%. According to EGM (1,1), China's revenue from inbound tourism is expected to reach \$57,319,000,000 (\$57.3bn) till 2026. The trend is increasing.

4.2.4 Forecasting outbound tourism expenditures. The fourth part (E) of the [Table 3](#) shows the actual data from 2006 to 2016 for expenditures incurred by the outbound tourism from China to abroad and predicted expenditures from 2017 to 2026 using grey models. All values of expenditures are in a hundred thousand (\$100,000). The results reveal, as far as expenditures made by outbound tourists from China is concerned, the model EGM's accuracy was 82.6%. According to EGM, expenditures made by outbound tourists from China are expected to reach \$3,833bn until 2026. The trend is increasing. These expenditures are the source of revenue for other countries that outbound tourists from China are choosing to be their travel destinations.

4.2.5 Improved forecasting through OPM. One advantage of OPM is that it can easily be integrated into a forecasting model without altering the core algorithm of the original model. The [Table 4](#) shows execution of O/P-EGM for outbound tourists' expenditure. The visual illustration of the actual data revealed that the recent data starting from 2013/2014 onward have bit different trend as compared to the preceding data. On this convenient basis, the actual data was divided into two parts, pessimistic data (actual data; historic dataset) and optimistic data (latest sub-set of pessimistic data).

Similarly, OPM-based forecasting for the other three cases (receipts from inbound tourists, inbound tourism and outbound tourism) can also be executed. The final results of this exercise for each case have been summarized in tabular form, as shown in the [Table 5](#).

Year	A (unit: 1,000 persons)		D (unit: 1,000 persons)		R (unit: 100,000 current US\$)		E (unit: 100,000 current US\$)	
	Actual data	EGM-based simulated data	Actual data	EGM-based simulated data	Actual data	EGM-based simulated data	Actual data	EGM-based simulated data
2006	49,913	49,913	34,524	34,524	339,490	339,490	243,220	243,220
2007	54,720	53,237	40,954	40,321	372,330	413,417	297,860	331,433
2008	53,049	53,771	45,844	46,448	408,430	420,588	361,570	425,580
2009	50,875	54,311	47,656	53,507	396,750	427,884	437,020	546,469
2010	55,664	54,856	57,386	61,639	458,140	435,306	548,800	701,699
2011	57,581	55,407	70,250	71,007	484,640	442,858	725,850	901,023
2012	57,725	55,963	83,182	81,798	500,280	450,540	1,019,770	1,156,966
2013	55,686	56,525	98,185	94,229	516,640	458,355	1,285,760	1,485,613
2014	55,622	57,092	116,593	108,549	440,440	466,306	2,273,440	1,907,615
2015	56,886	57,666	127,860	125,046	449,690	474,395	2,449,490	2,449,490
2016	59,270	58,244	135,130	144,050	444,320	482,624	3,145,290	3,145,290
2017		58,829		165,942		490,996		4,038,739
2018		59,420		191,161		499,513		5,185,979
2019		60,016		220,213		508,178		6,659,103
2020		60,619		253,680		516,993		8,550,682
2021		61,227		292,233		525,961		10,979,580
2022		61,842		336,645		535,085		14,098,430
2023		62,463		387,806		544,367		18,103,216
2024		63,090		446,743		553,809		23,245,599
2025		63,723		514,637		563,416		29,848,722
2026		64,363		592,848 (592.9 million)		573,190 (\$57.3 billion)		38,327,522
Parameters		$a = -0.0100$ $b = 52472.7881$ Coefficient = 52707.6773		$a = -0.01415$ $b = 32651.1784$ Coefficient = 35001.2452		$a = -0.0172$ $b = 404033.6231$ Coefficient = 406367.7948		$a = -0.250$ $b = 230912.957$ Coefficient = 258113.5325
MAPE (%)		2.64%		4.50%		7.67%		17.40%
Note(s): A: international tourism, number of <i>arrivals</i> (overnight visitors) D: international tourism, number of <i>départures</i> R: international tourism, <i>receipts</i> E: international tourism, <i>expenditures</i>								

Table 3.
EGM (1,1)-based
forecasting of tourists
and financials

Table 4.
Forecasting
international tourism,
expenditures (current
USD; \$100,000)
through O/P-EGM

Year	Pessimistic data (X)	Optimistic data (X')	X-based EGM	X'-based EGM	O/P-EGM
2006	243,220		243,220		243,220
2007	297,860		331,433		331,433.0
2008	361,570		425,580		425,579.6
2009	437,020		546,469		546,469.3
2010	548,800		701,699		701,698.9
2011	725,850		901,023		901,022.8
2012	1,019,770		1,156,966		1,156,966.4
2013	1,285,760	1,285,760	1,485,613	1,285,760	1,265,882
2014	2,273,440	2,273,440	1,907,615	2,294,541	1,957,505
2015	2,498,310	2,498,310	2,449,490	2,456,234	2,281,853
2016	2,611,290	2,611,290	3,145,290	2,629,322	2,685,080
2017			4,038,739	2,814,607	3,189,578
2018			5,185,979	3,012,949	3,824,380
2019			6,659,103	3,225,268	4,627,146
2020			8,550,682	3,452,549	5,646,745
2021			10,979,580	3,695,846	6,946,623
2022			14,098,430	3,956,288	8,609,170
2023			18,103,216	4,235,083	10,741,407
2024			23,245,599	4,533,525	13,482,380
2025			29,848,722	4,852,997	17,012,776
2026			38,327,522	5,194,982	21,567,420
Parameters			$a = -0.250$ $b = 230912.957$ Coefficient = 258113.5325 17.35%	$a = -0.0681$ $b = 2129745.7768$ Coefficient = 2143491.2070	
MAPE (%)					14.64%

Note(s): R: forecasting international tourism, receipts (current USD,\$100,000) through O/P-EGM
A: Forecasting international tourism, number of arrivals (1,000 persons) through O/P-EGM
D: Forecasting international tourism, number of departures (1,000 persons) through O/P-EGM

The OPM-based model reveals reasonable forecasting while considering both optimistic and pessimistic perspectives to predict future pattern of the data. For example, for the case of “international tourism–expenditure” using the EGM model, the position of O/P-EGM forecast against actual data (X)–based forecasts and optimistic data (X')–based forecasts can be seen from the [Figure 1](#).

4.2.6 Performance evaluation of the models with recently available data. The out-of-sample testing was performed on the 2017 and 2018 datasets, and the results from EGM, O/P-EGM and exponential regression analysis are shown in the [Table 6](#). MAPE (%) (2017, 2018) for the three models was (12.89%, 17.99%), (6.99%, 9.32%) and (19.29%, 21.77%). One can see that performance of O/P-EGM is finest among these models.

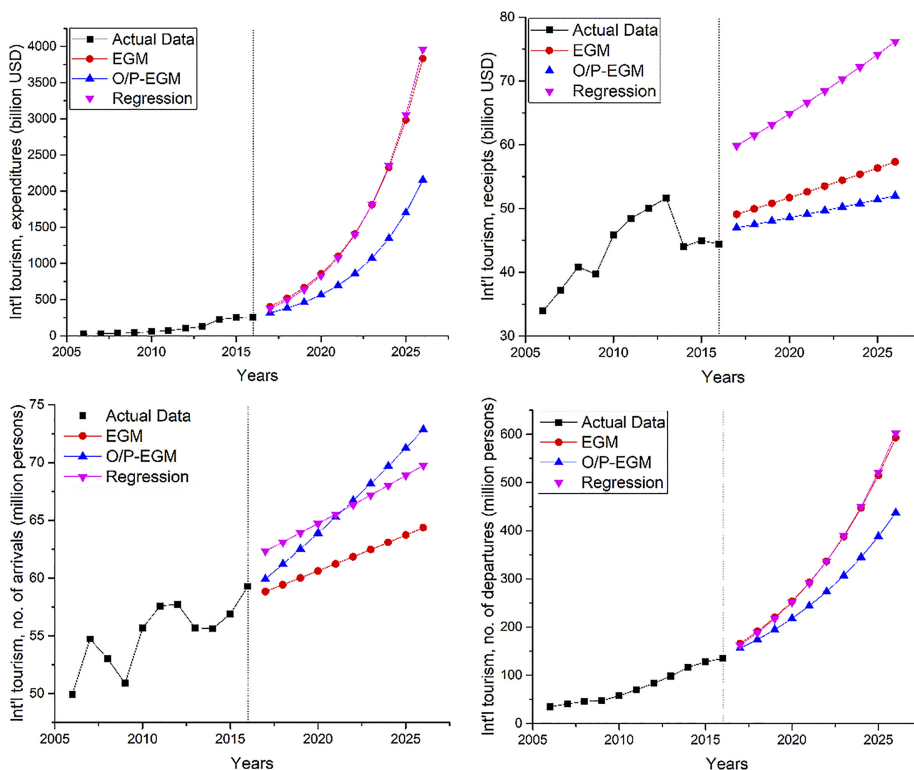
Along with the pilot testing, the [Table 6](#) reconfirms feasibility of OPM. Considering the Axiom – I in mind, it can be argued that decrease in the error measure (e.g. MAPE) for OPM-based forecast implies that the forecasted data is more explainable by the recent (optimistic) dataset than by the complete set of data (pessimistic). One can see from the [Table 6](#) that in all cases, OPM-based grey models' MAPE (%) values are lower that is signifying that the model is likely to produce reliable tourism-related predictions, which can be better explained by the recent data than by the entire data set. Thus, the findings suggest that the integration of OPM significantly improves the performance of grey prediction models, as can be seen from the [Figure 2](#) as well. In [Figure 2](#), the y-axis shows the four tourism indicators and the x-axis shows the forecast accuracy in percentage, deduced from MAPE (%) values.

Further, the study reveals that for tourism-related forecasting recent data contains more insight (concerning future) than older data because of dynamic and evolutionary nature of the tourism system; that's why OPM-based grey model generally performed better than the original model. The average annual growth rate of “International tourism, number of arrivals” was 1.84% from 2006 to 2016 and, according to O/P-EGM, is likely to increase to 2.2% from 2017 to 2026. The average annual growth rate of “International tourism, receipts” was 3.04% from 2006 to 2016 and is likely to reduce to 1.13% in the next ten years. The average annual growth rate of “International tourism, number of departures” was 14.8% from 2006 to 2016 and is likely to reduce to 12.1% in the next ten years. The average annual growth rate of “International tourism, expenditures” was 28% from 2006 to 2016 and is likely to reduce to 22.8% in next ten years, provided Chinese government's tourism policy or China's international relations do not see any drastic change.

5. Conclusion and recommendations

Tourism forecasting is a topic gaining exceeding attention in all those countries, who are either the primary source or recipient of tourists. Time-series models have shown to have strengths in forecasting tourism demand ([Morley, 2009](#)). Even GM (1,1) is one of the most popular grey forecasting models which uses time-series data as input to generate forecasts. The study shows when the proposed technique, OPM, was integrated into model Even GM (1,1), the mean absolute percentage error of the resultant model, O/P-EGM, was generally decreased that demonstrated increase in prediction accuracy. Thus, the study not only stresses the suitability of grey prediction models for tourism-related indicators but also shows (through a pilot case and real-world application) if OPM is integrated into grey models their performance can marginally be enhanced. Also, the study suggests for the compelling predictions of Chinese tourism-related indicators new data is more likely to be insightful than historical data. The recently available data (and out-of-sample testing) on the four cases confirmed the reliability of the predictions made by the proposed approach.

The benefits of the growth of the travel and tourism industry are at least three folds. It creates revenues for the country, its corporate sector and its domestic economy, enhances



Forecasting
China's
tourism
indicators

Figure 1.
Forecasting the four
tourism indicators
through EGM, O/P-
EGM and regression
models

employment opportunity in the country and creates a positive (hospitable) image of the country and its people abroad. According to the World Travel and Tourism Council (WTTC, 2017a; WTTC, 2017b), even though the US emerged as the strongest country in 2016 (followed by China), “a drop in inbound travel because of anti-foreign sentiment” is expected to reduce the growth rate of travel and tourism in the US, which faced a decline from 2.8% to 2.3% from 2016 to 2017. Being the strongest competitor of the US (in *absolute* terms of both direct and total contributions to GDP, travel and tourism investment and visitor exports, in 2016–2017) and the leader of the world (in terms of total contribution to employment) (WTTC, 2017a; WTTC, 2018), China has both opportunity and potential to attract foreign tourists through promotion of tourists-friendly sentiments and policies. The study outlines that Chinese tourism research is likely to seek more attention in future because of positive growth in all important indicators. Further, because of the rising wave of “protectionism” in competing economies, China has both resources and opportunities to make maximum gains by projecting itself as a tourist-friendly country.

The study suggests the future researchers to use OPM in other areas as well to enhance validity of the approach. Further, the impact of OPM's integration on forecasting techniques' long-term verses short-term prediction accuracy is an area of future research. Also, OPM-based grey models' suitability for other tourism indicators, and in other sectors of the economy, can also be gauged in future. Considering China to be one of the major travel and tourism countries of the world this study focussed itself to China, nevertheless future scholars can replicate the proposed framework in other countries as well with a view to comparatively

	International tourism expenditure (current US\$100,000)						International tourism receipts (current US\$100,000)					
	Actual data			Regression			Actual data			Regression		
	2017	2018	2019	2017	2018	2019	2017	2018	2019	2017	2018	2019
Variation	2,578,750	2,772,450	4,038,739	5,185,979	3,189,578	3,824,380	3,795,993	4,926,092	385,590	403,860	469,842	475,076
Variation	1,459,989	2,413,529	610,828	1,051,930	2,153,642	1,217,243	2,153,642	105,406	95,653	84,252	71,216	212,930
(%)	56.62%	87.05%	23.69%	37.94%	77.68%	47.20%	77.68%	27.34%	23.68%	21.85%	17.63%	55.22%
MAPE (%)	28.31%	43.53%	11.84%	18.97%	23.60%	38.84%	23.60%	13.67%	11.84%	10.93%	8.82%	27.61%
Outbound tourism from China (1,000 persons)												
	Actual data			Regression			Actual data			Regression		
	2017	2018	2019	2017	2018	2019	2017	2018	2019	2017	2018	2019
	143,035	149,720	165,942	191,161	155,992	174,119	72,616	83,947	60,740	62,900	59,420	61,207
Variation	22,907	41,441	12,957	24,399	70,419	65,773	70,419	3,15%	5.53%	1.35%	2.69%	6.23%
Variation	16.01%	27.68%	9.06%	16.30%	49.23%	43.93%	49.23%	3.15%	5.53%	1.35%	2.69%	6.23%
(%)	16.01%	27.68%	9.06%	16.30%	49.23%	43.93%	49.23%	3.15%	5.53%	1.35%	2.69%	6.23%
MAPE (%)	8.01%	13.84%	4.53%	8.15%	24.62%	21.97%	24.62%	1.57%	2.77%	0.67%	1.35%	1.31%
Inbound tourism to China (1,000 persons)												
	Actual data			Regression			Actual data			Regression		
	2017	2018	2019	2017	2018	2019	2017	2018	2019	2017	2018	2019
	143,035	149,720	165,942	191,161	155,992	174,119	72,616	83,947	60,740	62,900	59,420	61,207
Variation	22,907	41,441	12,957	24,399	70,419	65,773	70,419	3,15%	5.53%	1.35%	2.69%	6.23%
Variation	16.01%	27.68%	9.06%	16.30%	49.23%	43.93%	49.23%	3.15%	5.53%	1.35%	2.69%	6.23%
(%)	16.01%	27.68%	9.06%	16.30%	49.23%	43.93%	49.23%	3.15%	5.53%	1.35%	2.69%	6.23%
MAPE (%)	8.01%	13.84%	4.53%	8.15%	24.62%	21.97%	24.62%	1.57%	2.77%	0.67%	1.35%	1.31%

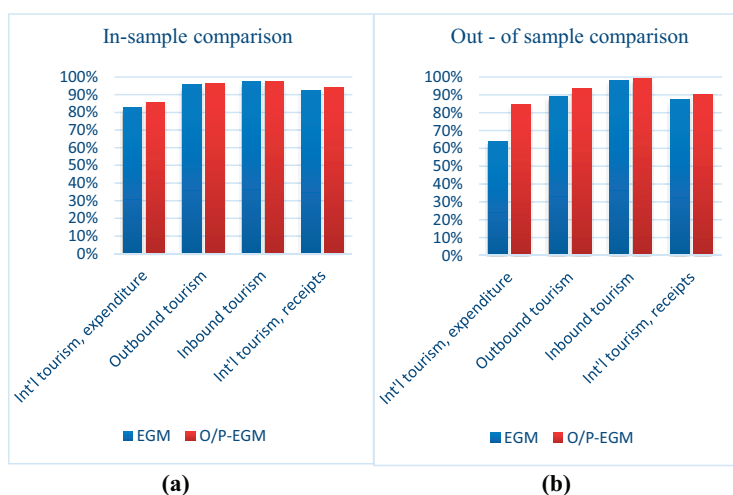


Figure 2.
Forecast accuracy (%)
of the models (a) EGM
and (b) O/P-EGM

analyse different economies while considering the increasing role of tourism in supporting several countries' national economies and absorbing their surplus labour. For example, rapidly developing countries that are going through uneven distribution of development despite abundance of natural or historic resources can particularly benefit from these proposals.

No development is free from shortcomings. OPM is a modest effort to make a forecasting more accurate when robust forecasting methods cannot be applied because of limited resources, e.g. when time constraint or higher computation cost of a forecasting method prevents a decision-maker from using it, and thus a convenient methodology is needed. Thus, OPM provides decision-makers with a convenient technique to saturate accuracy of forecasting models they are already familiar with. However, convenience comes at a cost. OPM-based results should only be trusted when the Axiom – I is satisfying. Further, the value of the coefficient α should be determined based on the guidelines discussed in this paper, and its value should be altered when needed to bring forecast nearer to reality. In the future, a robust algorithm to estimate the optimum value of α can be developed. An approach that can inconsistently distribute the value of α considering the recency of data at each point of time can better serve the purpose. Also, how the integration of OPM in other forecasting methods would influence accuracy of those methods is yet to be seen. Moreover, gauging short-term and long-term impacts of the COVID-19 pandemic on travel and tourism industries is a topic requiring systematic investigations in future.

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